

ReSolve: Verified Mechanical CAD Edits and Linked Vector Drawings

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Paper ID Draft

Abstract

001 Editing a mechanical drawing is useful only when the
002 underlying part changes as requested and the delivered
003 views remain consistent. We study a CAD-first work-
004 flow that predicts a validated program patch, executes
005 the edited CadQuery model, and exports linked SVG
006 views. Our main experiment uses published Bench-
007 CAD edit pairs with a local family- and template-
008 component-disjoint train/validation/test split. A fresh
009 1.5B QLoRA model is trained on 492 retained edits
010 and evaluated on all 126 retained held-out cases, us-
011 ing source-program equality and the author’s released
012 STEP solids. The adapter yields 77/126 applicable
013 patches and 12 exact target programs; 12 of 125 valid
014 released STEP references are matched strictly. A sub-
015 sequent three-epoch H100 LoRA run on the same re-
016 tained split produces 61/126 applicable patches and
017 15/125 strict solid matches. We report executable
018 solids and volume agreement separately, with a format-
019 relaxed base-model control. A smaller, explicitly local-
020 ized pilot provides 150 verified edit pairs and paired
021 drawing sheets. Separate deterministic truss and plate
022 controls demonstrate axial FEM equilibrium and clear-
023 ance violations; those results are not attributed to the
024 learned CAD model. A separate Colab 3B adapter
025 trained on 2,540 edits from two online sources obtains
026 39/252 strict held-out matches; four of six illustrative
027 new-model cases pass a combined geometry/FEM di-
028 agnostic. The study measures code-conditioned edit-
029 ing and linked view generation, not drawing percep-
030 tion or general engineering safety. The release lacks
031 physical operating conditions; a separate, explicitly
032 scenario-defined 3D elasticity audit checks six illus-
033 trated source/target/prediction triples after training.
034

1. Introduction

A request to remove a bracket flange, enlarge a bore or
add a shaft slot must change more than a caption. The
edited part should preserve unrelated features, remain
executable, and produce coherent views. If the request
concerns structural performance, it additionally needs
material, units, supports, loads and a declared phys-
ical model. Shape validity alone cannot supply those
assumptions.

This paper focuses on mechanical CAD edits and
their linked vector drawings. Figure 1 illustrates two
published reference pairs: removing a flange while re-
taining the twisted web and first-flange holes, and
adding an obround blind slot to a stepped shaft. These
require compound three-dimensional geometry rather
than a symbol substitution. The figure shows released
references, not model predictions.

We implement a CAD-first edit-execute-check
route. A language model reads an executable pro-
gram and the user’s instruction and returns a source-
line patch. A validating executor applies the patch to
the original program. CadQuery builds the resulting
solid; orthographic and isometric SVG views are ex-
ported from that solid. Independent scoring compares
the executed result with the author’s STEP reference.
A malformed patch, invalid solid and valid but incor-
rect edit are distinct outcomes.

Our main data comes from an online release rather
than only locally manufactured requests: BenchCAD’s
published edit pairs [14]. Because its edit subset is
originally a held-out benchmark, using it for training
changes the protocol. We explicitly construct a local
split before training, hold out complete family/tem-
plate components and start from fresh base weights.
We do not report an official BenchCAD leaderboard
result.

Contributions and scope.

- A reproducible local training/evaluation protocol

073	over published mechanical edit pairs, with pinned	
074	provenance, family/template isolation, disclosed to-	
075	ken exclusions and every retained test case scored.	
076	• A validated CAD-program patch interface and de-	
077	terministic linked SVG export, evaluated with both	
078	program and executed-solid agreement. The learned	
079	model receives code; it is not a drawing-perception	
080	model.	
081	• A separate 150-pair hard-part pilot with be-	
082	fore/after STEP solids and four-view drawing	
083	sheets, plus a transport control showing why patch-	
084	format gains must not be equated with engineering	
085	reasoning.	
086	• Explicit separation of mechanical geometry checks,	
087	deterministic structural FEM equilibrium and plate	
088	clearance rules. No mechanical safety or broad	
089	method-novelty claim follows from these small,	
090	single-seed experiments.	
091	2. Related work	
092	Executable CAD and editing. DeepCAD [12] and	
093	Text2CAD [5] model parametric construction histo-	
094	ries. CAD-Editor [13] studies instruction-based se-	
095	quence editing. BenchCAD [14] supplies executable	
096	mechanical programs and published edit pairs, while	
097	neuralCAD-Edit [8] evaluates expert-requested ed-	
098	its. Our experiment reuses released supervision; a	
099	program-patch interface and execution check are im-	
100	plementation choices, not evidence that CAD editing	
101	itself is new.	
102	Drawings and vector graphics. Drawing2CAD [9]	
103	links vector views to CAD histories. DeepSVG [1] and	
104	StarVector [10] model vector graphics, and SVGE-	
105	ditBench [7] benchmarks instruction-based SVG edits.	
106	Our learned experiment conditions on CAD code and	
107	exports SVG after execution. It therefore does not	
108	establish SVG-to-CAD extraction, image understand-	
109	ing, or robust annotation recovery from manufacturing	
110	sheets.	
111	Physical evaluation. FEM-Bench [6] tests scientific	
112	code generation. CADEngBench [11] evaluates func-	
113	tional CAD edits and matched linear-static FEA. Ge-	
114	ometry and FEA are complementary: the latter re-	
115	quires an analysis contract. We retain an explicitly	
116	specified axial-truss control but do not fabricate loads	
117	or materials for BenchCAD parts. A public runnable	
118	CADEngBench release was not verified in this work, so	
119	its reported FEA results are related work rather than	
120	our reproduced benchmark.	
3.	Mechanical edit-execute-check workflow	121
	Figure 2 illustrates a functional flange edit: replace	122
	four mounting holes with six equally spaced holes while	123
	retaining hole diameter and bolt-circle radius. The ac-	124
	tual saved patch changes only the polar-pattern count	125
	from 4 to 6; geometry and FEM checks follow execu-	126
	tion. The trained CAD editor architecture is shown in	127
	Fig. 5.	128
	3.1. Program and patch contract	129
	Let C be a CadQuery program and u an edit instruc-	130
	tion. The model predicts a patch p , and the executor	131
	produces $\hat{C} = \mathcal{E}(C, p)$. Each patch operation contains	132
	a zero-based original line index, a deletion count and	133
	a list of replacement lines. All coordinates refer to the	134
	original source, not an intermediate revision. Opera-	135
	tions must be ordered, non-overlapping, within bounds	136
	and well typed. Parsing the executed program provides	137
	a second syntax check.	138
	Training targets are line diffs between the published	139
	source and edited programs after removing comments	140
	and empty lines. Comment normalization avoids su-	141
	pervising changes to the upstream parameter-comment	142
	block as geometric edits. A round-trip check requires	143
	that applying every target patch reproduces its nor-	144
	malized target exactly. The online instructions are	145
	preserved; source-line answers are not added to the	146
	prompt.	147
	3.2. Execution and vector views	148
	CadQuery executes the patched program to a	149
	boundary-representation solid $\hat{B}$. The scorer requires	150
	valid geometry, at least one solid and positive volume.	151
	Downloaded programs run through a restricted execu-	152
	tor with permitted imports, no private-attribute access	153
	or file-I/O builtins, and a 45-second per-case subprocess	154
	budget. This restriction is defense in depth, not a	155
	general Python sandbox. Unsupported constructs, in-	156
	valid programs and timeouts are recorded separately.	157
	For the verified drawing pilot, front, top, right	158
	and isometric SVG projections of both original and	159
	edited solids are saved alongside STEP exports. A be-	160
	fore/after sheet binds these views to the request and ge-	161
	ometry measurements. The projections share a three-	162
	dimensional source; they are not independently gen-	163
	erated views. They are CAD-derived line drawings	164
	with edit annotations, not fully dimensioned produc-	165
	tion blueprints. Automatic projection scaling can differ	166
	between views, so rendered pixel size is not a physical	167
	measurement. Figure 7 shows sixteen such projections	168
	of two source/edited pairs from the completed 3B run.	169

Published BenchCAD source / reference edit pairs

twisted bracket | Remove the second flange; retain the twisted web and first-flange holes.

SOURCE

RELEASED TARGET



stepped shaft | Add a centered blind obround slot to the upper shaft face.

SOURCE

RELEASED TARGET



Reference illustrations only. No stress or FEM result is inferred from these shapes. BenchCAD, CC-BY-4.0.

Figure 1. Published mechanical CAD edit pairs held out in the local protocol. Top: remove the perpendicular end flange while preserving the twisted web and the first-flange holes. Bottom: add a centered blind obround slot to the upper cylindrical shaft section. These are source/released-target illustrations, not learned model predictions. Both views are exported from solids; shape appearance supplies no stress or FEM result; the separate FEM audit concerns the six actual predictions in the supplement.

3.3. Independent solid scoring

The online test reference is the author’s released STEP solid B^* , pinned by content hash. We report volumetric intersection-over-union,

$$\text{IoU}(\hat{B}, B^*) = \frac{V(\hat{B} \cap B^*)}{V(\hat{B}) + V(B^*) - V(\hat{B} \cap B^*)}. \quad (1)$$

Our local thresholds are 0.95 for broad volume agreement and 0.99999 for strict agreement. Neither is presented as an official BenchCAD acceptance rule. Exact normalized-program AST equality is complementary: IoU may overlook a small but important feature, and multiple programs can produce the same shape. The current scorer does not certify feature-specific tolerances, preserved constraints, assembly joints, dimensions, standards compliance or manufacturing suitability.

Verification is post-edit evaluation. The QLoRA loss is supervised token prediction, not FEM or solver feedback; the experiment does not evaluate iterative repair after a failed check.

4. Online dataset and split protocol

Published pairs. We use the `edit-bench` configuration of BenchCAD/BenchCAD, pinned at revision `5919f578ab09ec283603a082fab07c7639ab56eb`. It contains 748 instruction/source/target pairs with original and target STEP data [14]. The dataset declares CC-BY-4.0. Our downloaded parquet checksum and source-record identifiers are retained. Pipe, duct, fitting, manifold, valve, hose, resistor, circuit and PCB family names are excluded, removing 38 records and leaving 710 mechanical edit pairs.

Isolation before training. A complete mechanical family is one initial group. Families whose source or target ASTs share a template after masking numeric literals are joined into a component. Hash-ranked components are assigned approximately 70/10/20 to training, validation and test before normalization, patches or augmentation. This gives 533/50/127 pairs. Family and component identities are disjoint across splits; no target from a held-out family enters fine-tuning. Template isolation is conservative but does not prove all cross-dataset geometric ancestry is disjoint. The fresh run

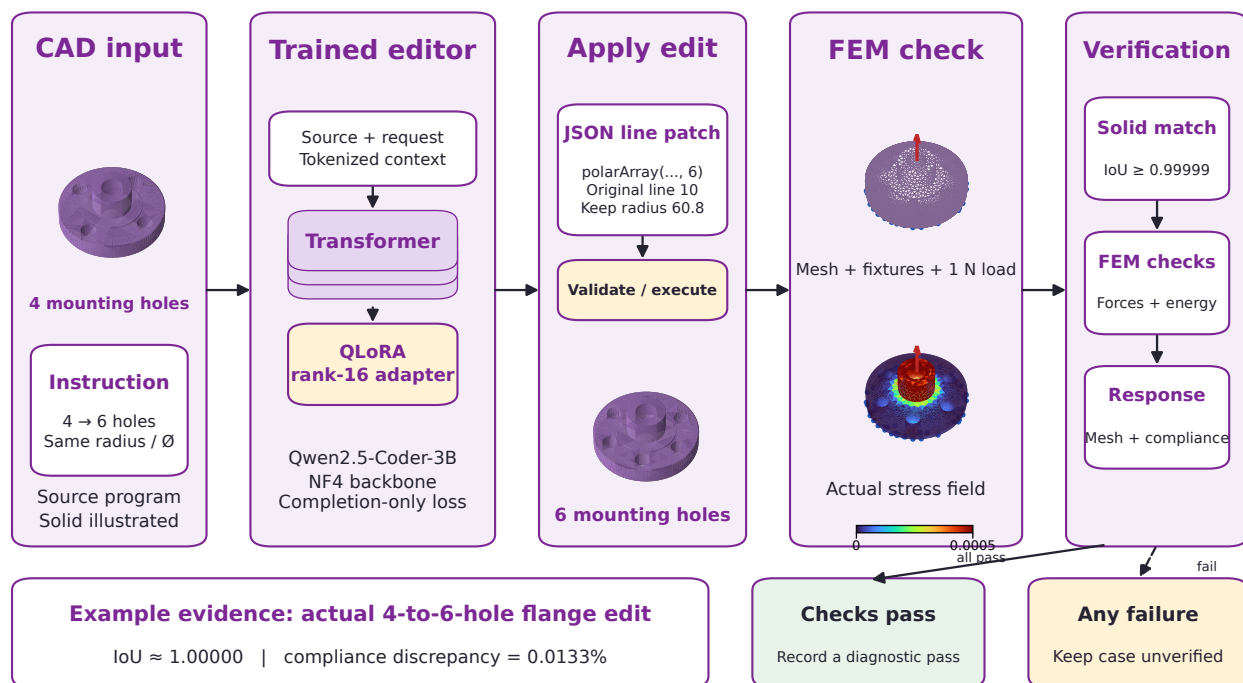


Figure 2. Illustrated mechanical CAD editing and FEM verification architecture. The source program and instruction feed the trained CAD editor, which returns an original-coordinate JSON line patch. The source solid is shown for illustration. Schema checks and execution produce the actual edited solid. The flange shown here is the final 3B model output for a four-to-six mounting-hole edit, retaining the released hole diameter and bolt-circle radius: its saved tetrahedral mesh, fixed nodes (blue), load direction (red), and von Mises field illustrate the disclosed synthetic 1 N test. Stress colors are diagnostic, clipped at p95 in MPa. Independent solid agreement, FEM equilibrium/energy, mesh sensitivity and target compliance determine the recorded outcome. Failures remain unverified; FEM does not supervise CAD token training.

211 does not load the earlier mechanical adapter.

212 Context-limited coverage. We retain a pair only when
213 its full chat, including target, fits 3,072 tokens and
214 its target patch fits 512 tokens. The rule is applied
215 once before baseline evaluation and training. It re-
216 tains 492/44/126 train/validation/test pairs, excluding
217 48 records across all splits. No example is silently truncated.
218 The one excluded test pair and all excluded
219 training examples remain identified in the run manifest.
220 Results apply to the retained slice rather than
221 the entire release.

222 Protocol change. The author labels this edit subset
223 a held-out benchmark. We deliberately re-partition
224 it into local training and evaluation data to satisfy
225 the present study’s training objective. This invalidates
226 claims of official benchmark-test generalization
227 or direct leaderboard comparability. The local held-out
228 components remain unused for optimization and
229 checkpoint selection.

Reference-quality limits. Readback finds one invalid
rivet STEP among the 126 retained test references.
We also identify a concrete instruction/target conflict:
topup_stepped_shaft_top_slot_rev requests cutting a blind slot,
but its source already has the slot and its target removes the operation.
Published labels are retained without silently repairing them;
this case and its source diff are recorded. The experiment
measures agreement with released targets, not certified compliance
with every natural-language request. We do not infer that every
reverse-edit record has this defect.

Earlier drawing pilot. A separate sample of 2,000
BenchCAD code-generation parts yielded 150 verified
local numeric-edit pairs from 76 hard parts and 40 families,
partitioned 108/22/20 by family. It requires at least eight CAD
calls and compound geometry, and excludes external edit-program
templates after masking literals. Chamfer, hole, extrusion and
fillet edits contribute 62, 48, 24 and 16 pairs. The sample exports

904 referenced SVG views and 150 before/after sheets.
Its instructions disclose the edit line and numeric answer; it is a transport feasibility pilot, not the main online benchmark.

5. Online editing experiments

Training. We start from a fresh 4-bit Qwen2.5-Coder-1.5B-Instruct conversion [4] and train a QLoRA adapter [3] using MLX on an Apple M5 with 16 GB memory. The model conversion revision is b3252a2f97102b1fb1571fec2c9b27219a8536be. Seed 17, batch size one, learning rate 10^{-4} and the last eight transformer blocks are fixed. Adapters have rank 8, scale 20, no dropout, and cover attention and MLP linear projections in those blocks. Prompt tokens are masked from the loss. We run 492 optimizer steps, corresponding to one pass of the retained training rows, and use the final checkpoint. Validation is logged, not used to select among test-informed alternatives. Saved adapters are reloaded before evaluation.

Evaluation and metrics. Base and trained models receive the same original programs, published instructions and patch contract. Greedy decoding is capped at 512 new tokens. All 126 retained test records are evaluated in the same fixed order. Applicability requires a valid patch and syntactically valid program; AST equality requires the executed program to match the published target’s normalized syntax. Solid scoring executes the actual prediction and compares it with the released STEP reference. No gold edits repair a model prediction.

Transport control. The base model frequently emits a bare list of edits rather than the required object containing `edits`. We therefore add a deterministic diagnostic that wraps a syntactically valid bare list while preserving every operation. It makes 56 of 126 base predictions applicable, but none matches the target AST. This control distinguishes one schema mismatch from the remaining content and coordinate errors; it does not rewrite geometry, infer missing operations or correct indexing.

Results and operational coverage. The trained model produces 77/126 applicable patches and 12/126 exact target ASTs. Of 125 independently valid STEP references, 39 predictions build valid solids, 18 reach $\text{IoU} \geq 0.95$, and 12 reach strict $\text{IoU} \geq 0.99999$ (9.6%). The bare-list base control builds 3 solids, with 1 broad and 0 strict matches. Thus the experiment exposes a substantial remaining gap: most held-out edits fail to

produce the reference geometry, particularly beyond literal-parameter changes. Independent reference read-back finds 125/126 valid target solids. One released rivet reference is invalid and is excluded from the geometric success denominator, while all task outputs remain reported. For the trained run, 39 cases are geometrically scored, 37 have execution errors and 49 have invalid patches; there are 0 timeouts and 0 Boolean-comparison errors. The observed code failures include invalid feature operations and damaged source coordinates; they are distinct from valid solids with low IoU. We keep base and trained outputs, patch errors, executed-solid measurements, reference hashes and category breakdowns. One seed and one model size cannot establish a robust scaling law or broad superiority over CAD agents. No official benchmark claim or statistical population failure rate is inferred from this local slice.

Additional H100 run. After the local experiment, available server compute supports a separately reported CUDA run. A fresh unquantized BF16 Qwen2.5-Coder-1.5B-Instruct model is trained on the identical 492 retained training rows for three epochs (1,476 updates) on one NVIDIA H100 NVL. PEFT adapters use rank 8, alpha 160 (scale 20), no dropout and the same seven attention/MLP projections in the last eight blocks. Seed, batch size, learning rate, prompt masking, context limit and greedy generation limit remain unchanged. We reload the final adapter and evaluate all 126 cases; no test-based checkpoint selection is performed. Different precision, software implementations and epoch counts prevent interpreting the comparison as an isolated hardware effect or an independent preregistered confirmation.

The final CUDA adapter yields 61/126 applicable patches and 16/126 target AST matches. Against 125 valid released STEP solids, 36 predictions execute, 19 meet broad IoU and 15 meet strict IoU (12.0%). There are 65 invalid patches, 24 execution errors, 0 timeouts and 0 comparison errors; the same invalid released reference is excluded from the geometry denominator.

Localized drawing pilot. The earlier 100-step adapter retained 92 training, 22 validation and 12 test examples after context filtering. All 12 held-out exact patches execute and match their references. Under a unique-substring line parser the base model already matches 11/12 solids. That result mainly improves transport formatting and cannot support a claim of advanced CAD reasoning. The online experiment uses published instructions without line hints, a fresh adapter and its own held-out families.

System	Applicable	Target AST	IoU ≥ 0.95	IoU ≥ 0.99999
Base, strict interface	0	0	0	0
Base, bare-list transport control	56	0	1	0
Fresh online-pair QLoRA	77	12	18	12

Table 1. Local online-pair benchmark: counts out of 126. All systems use the same family/template holdout. The transport control only wraps a bare JSON edit list into the required object, without changing coordinates or content. Timeouts and comparison errors remain separately disclosed; they are not measured incorrect geometry. This is a local split, not an official BenchCAD score.

Category	Cases	Target AST	Strict match
T1	31	11	11
T2	18	1	1
T3	18	0	0
T4	34	0	0
T5	25	0	0

Table 2. Trained results by published edit category. Category IDs are retained from the source; counts are not pooled across different evaluation contracts.

Completed multi-source Colab extension. A separate fresh 3B QLoRA adapter trains on 2,540 paired edits from BenchCAD and CAD-Editor. On 252 held-out tasks it yields 173 applicable patches, 139 executed solids and 39 strict geometry matches (15.48%), versus one strict match for its fresh base. Appendix D reports the full protocol, source-specific results and new-adaptor FEM audit. This extension uses a different model and corpus; its counts are not pooled with Table 3.

6. Separate structural and clearance controls

The full CAD benchmark above checks mechanical shape. A separate six-case FEM audit is described in the supplement. We retain a separately identified deterministic SVG route to show what physical or manufacturing checks require when a drawing does supply a contract. These synthetic controls are not training data or accuracy evidence for the learned BenchCAD model.

Axial trusses. An importer reads supported untagged truss geometry, section dimensions, material notes, nodal loads and supports. A bounded request interpreter generates validated SVG tree patches. Re-importing the actual patched SVG determines member lengths, orientations and axial areas. Linear-elastic pin-jointed elements use EA/L stiffness; assembled displacements, stresses and reactions are recomputed. The numerical verdict checks equilibrium under small-displacement and nodal-load assumptions, not buckling, fatigue, joints or material strength.

Plate clearances. A separate importer recovers a rectangular boundary and circular-hole centres, diameters, thickness and stated limits. It computes overlap, edge distance and ligament. A faithful hole-enlargement request can reduce clearance below the printed minimum. A geometric violation is an engineering consequence of a correctly executed request, not necessarily an edit error; no continuum plate stress follows from a clearance score.

For example, increasing all truss member depths from 20 to 40 mm changes the recovered area and reduces peak axial stress from 11.861 to 5.930 MPa under the same load. Enlarging plate holes from 10 to 16 mm at fixed centres 20 mm from a boundary reduces edge clearance from 15 to 12 mm and fails a 15 mm rule. These are distinct checks with explicitly stated assumptions. The saved audit and replayed source/edited SVGs remain separate from the new online training run.

When FEM is applicable. A stress, stiffness, buckling, thermal or contact request needs physical units, constitutive properties, loads and boundary conditions, followed by solver and mesh checks. The inspected BenchCAD edit pairs do not provide this analysis contract. We retain not evaluated for the full released benchmark and distinguish the added six-case, synthetic-load FEM audit from validation under actual operating conditions.

Verification coverage. All six illustrated H100 cases are actual learned CAD program edits, followed by execution and comparison with released STEP targets. Each is now submitted to a separate 3D FEM response and mesh audit under the documented synthetic contract. The online benchmark and localized mechanical pilot check geometry; the plate controls check clearances. The six-case supplement solves 3D linear-elastic FEM problems, while the separate truss controls solve axial FEM equilibrium. Geometry agreement does not establish stress, stiffness or safety. FEM for a mechanical part would require an explicitly specified material,

System	Applicable	Target AST	IoU ≥ 0.95	IoU ≥ 0.99999
CUDA base	0	0	0	0
CUDA base, list wrapper	47	0	0	0
CUDA final LoRA	61	16	19	15

Table 3. Additional three-epoch H100 experiment. Counts cover the same 126 retained tasks and 125 valid references, using the unchanged local geometry thresholds. Both base and final model are decoded with the CUDA implementation; the wrapper control preserves predicted edit contents.

Route	Edits	Outcome
Axial trusses	20,008	Equilibrium passes
Perforated plates	19,992	699 clearance violations

Table 4. Historical structural pipeline audit. All 40,000 supported edits match their stored target trees. These counts describe a deterministic synthetic pipeline, not a learned CAD benchmark. The plate violations are faithful edits flagged by the geometric checker.

unit system, load case, supports and a validated mesh/-solver setup, which are absent from these released edit pairs.

7. Limitations and conclusion

This is a small, single-seed code-conditioned CAD editing study. The online supervision comes from a synthetic mechanical benchmark re-partitioned locally; family/template checks do not prove complete geometric-ancestry isolation. A documented instruction/target conflict and one invalid STEP reference show that published labels require further auditing. Context filtering removes some of the longest parts and edit patches. Predicted code is limited by the executor and its time budget, and the current comparison checks solid volume agreement rather than feature-specific engineering tolerances. Our SVG views are deterministic projections; no learned drawing perception, annotation binding or production-sheet reconstruction is evaluated.

The structural controls are deterministic and use bounded grammars. Their equilibrium and clearance results cannot be transferred to arbitrary CAD shapes or interpreted as general safety certification. The full mechanical benchmark has no author-specified physical contract. We add a separate post-hoc 3D FEM audit of six illustrated edits under documented synthetic fixtures and loads; its response checks do not establish operating safety.

The completed two-source Colab extension reaches 39/252 strict matches; its four verified illustrative prediction/target pairs do not imply full-benchmark FEM

coverage. Invalid and incomplete examples remain unverified.

The completed workflow links published CAD edit supervision, validated program patches, executed solids and vector views. By reporting schema adherence, target syntax, actual solid agreement and operational failures separately, it makes the supported outcome reviewable. Stronger evidence requires larger contexts, multiple seeds and models, feature-preservation metrics, genuine drawing-conditioned editing and physical tasks with independently specified analysis contracts.

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519 A. Reproduction and evidence contracts

520 Online source. The `edit-bench` parquet is down-
521 loaded from [https://huggingface.co/datasets/](https://huggingface.co/datasets/BenchCAD/BenchCAD)
522 [BenchCAD/BenchCAD](https://huggingface.co/datasets/BenchCAD/BenchCAD) at the revision stated in Sec. 4.
523 The original published split name is `edit_bench`. Its
524 748 rows are not silently treated as 748 independent
525 source shapes; grouping uses all original and edited
526 programs before split assignment. Numeric literals are
527 masked for template linkage; all records of a linked
528 component retain one split.

```
529 {"edits": [  
530   {"start": 5, "delete": 1,  
531    "insert": ["    .hole(6.0)"]}]  
532 ]}
```

533 The zero-based line coordinate addresses the original
534 code. A replacement list is required even for one line.
535 Insertion uses zero deletions. The executor rejects over-
536 lapping operations and embedded newline characters.
537 This example describes the interface, not an online gold
538 answer.
539

540 Saved artifacts. The online dataset manifest pins the
541 parquet checksum, split checksums, source IDs, cat-
542 egories and reference STEP hashes. The run mani-
543 fest pins model revision, package versions, seed, opti-
544 mization command, retained rows and omissions. The
545 baseline and reloaded final adapter use identical test
546 prompts. Predictions remain unmodified; the bare-list
547 transport control is saved as a separate diagnostic. Ge-
548 ometry scores include every retained case, including
549 invalid patches, code errors, unsupported constructs,
550 timeouts and Boolean-comparison errors.

551 Evidence separation. The 150-pair localized pilot,
552 710-pair online protocol and historical 40,000-edit
553 structural audit are different datasets and contracts.
554 Their results are never pooled. The localized pilot's
555 family test contains only 12 context-retained examples
556 and gives line/number hints. The online experiment
557 is trained afresh with original natural-language re-
558 quests and evaluates 126 context-retained cases. Struc-
559 tural FEM is only established for specified pin-jointed
560 trusses.

561 Validation. Regression checks test multi-operation
562 original-coordinate execution, insertion/deletion, mal-
563 formed indexing, overlap rejection and comment
564 preservation. Dataset checks require train/valida-
565 tion/test family and template-component disjointness,
566 exact target-patch replay and pinned split hashes. Fi-
567 nal adapters are reloaded for generation and checked

for finite tensors. Reference STEP readback and all
output geometry scores are retained. Source/refer-
ence illustrations are labeled as such and do not depict
model predictions.

B. Held-out model edit examples

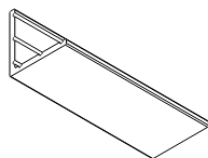
Figures 3 and 4 show six actual predictions from the
final three-epoch H100 adapter, alongside the input
solid and released STEP target. These are illustrative
cases selected after evaluation, not a random sample
or additional test set; all 126 evaluated cases remain
included in the reported aggregate results. We select
three strict matches and three executable but geometri-
cally incorrect edits to expose both supported behavior
and remaining limitations. Each view is independently
fitted to its panel, so apparent drawing size is not a
dimensional comparison. Instructions and measured
volume IoU accompany every row. These shape plates
alone provide no FEM evidence; the separate scenario-
defined audit below supplies the material, loading, sup-
port and solver records.

Held-out H100 CAD edits: strict matches

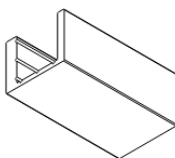
mounting angle box x f130 | T1 | IoU 1.00000

Increase the box X dimension from 33.0 to 42.9.

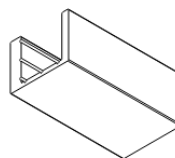
SOURCE



RELEASED TARGET



H100 PREDICTION



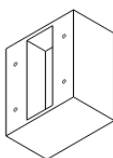
locator block box y f130 | T1 | IoU 1.00000

Increase the box Y dimension from 82.2 to 106.86.

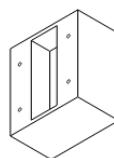
SOURCE



RELEASED TARGET



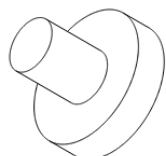
H100 PREDICTION



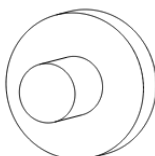
t3medplus stepped shaft rot diag60 | T2 | IoU 1.00000

Rotate the entire part by 60 degrees about the axis (1, 0, 1).

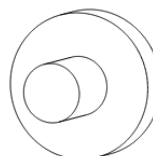
SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. FEM audit reported separately.

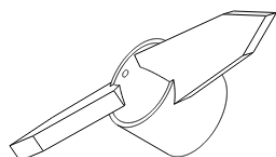
Figure 3. Three successful held-out H100 edits. Rows show a mounting-angle dimension change, locator-block width change and a 60-degree stepped-shaft rotation. Columns contain the source, author’s released target and actual final-adaptor prediction. Each meets strict volume IoU ≥ 0.99999 . These examples demonstrate reference agreement within the disclosed code-conditioned protocol; they do not certify general engineering correctness.

Held-out H100 CAD edits: partial / failed edits

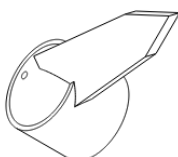
propeller bore widen compute | T4 | IoU 0.63155

Remove one of the two propeller blades; keep the hub, spinner, and the remaining blade.

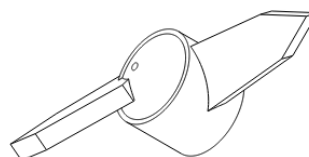
SOURCE



RELEASED TARGET



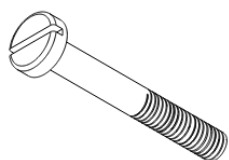
H100 PREDICTION



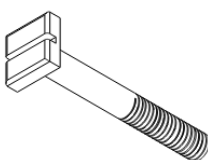
t5hard pan head screw U a | T5 | IoU 0.58449

Replace the round pan head of the screw with a square head of the same across-flats width and the same thickness, keeping the threaded shaft, the screwdriver slot, and the rounded top edge unchanged.

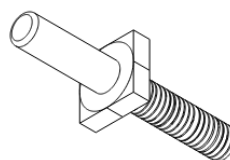
SOURCE



RELEASED TARGET



H100 PREDICTION



t5hard round flange V a | T5 | IoU 0.92018

Change the ring of 4 polar mounting through-passages on the flange face to a ring of 6 mounting through-passages of the same diameter, evenly spaced around the central axis at the same radius.

SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. FEM audit reported separately.

Figure 4. Three executable but incorrect held-out H100 edits. Removing a propeller blade, replacing a screw head and changing a flange mounting-hole pattern yield valid solids but fail the strict reference threshold. Their volume IoUs are approximately 0.632, 0.584 and 0.920. Source, released target and model output remain separate; no reference operations repair these predictions. Small feature errors can remain important even when bulk volume overlap is high.

C. Scenario-defined FEM audit of mechanical edits

Figure 6 gives a visual walkthrough using two actual final 3B edited geometries; its mesh, displacement and stress panels are linked to saved solver fields.

Scope and physical contract. We attempt FEM on every source, released target and actual prediction in the six illustrated H100 cases: 18 configurations, independently of model training. The released dataset supplies no operating load case. We explicitly assume millimetres, newtons and MPa, a homogeneous isotropic material with $E = 210,000$ MPa and $\nu = 0.3$, and a 1 N resultant tensile test load. These are synthetic diagnostic conditions, not recovered material properties or certified operating conditions. All displacements are fixed at exterior mesh nodes in the lower 5% of the shape’s projected span along the analysis axis. A uniform vector traction is integrated over exterior triangles whose centroids occupy the upper 5%; its area-weighted nodal forces sum to 1 N. The same geometric fixture rule is applied separately to each shape, rather than claiming identical world-coordinate contact patches. The shaft’s fixture and load axis co-rotate with its requested 60-degree pose edit. Every case contract and STEP hash is saved before observing FEM responses.

Discretization and numerical checks. Gmsh [2] meshes the actual STEP geometry with first-order tetrahedra. A SciPy sparse implementation solves small-strain 3D isotropic elasticity,

$$\epsilon = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \quad \sigma = \lambda \operatorname{tr}(\epsilon) \mathbf{I} + 2\mu \epsilon. \quad (2)$$

The element stiffness is $K_e = V_e B_e^T D B_e$, with engineering shear strains. Dirichlet elimination solves the free-degree system. Checks require finite displacements, anchored connected components, positive strain energy, free residual, force and moment balance, and the identity $2U = \mathbf{f}^T \mathbf{u}$, each normalized numerical error below 10^{-6} . An analytical uniform-tension cube agrees to numerical precision; load/modulus scaling and co-rotated-fixture tests also pass.

Mesh sensitivity and response comparison. We report compliance $C = \mathbf{f}^T \mathbf{u}$ in Nmm. Global mesh sizes start at one twelfth and one eighteenth of the case’s maximum bounding-box extent, with further refinement if the relative compliance change exceeds 5%. The propeller source requires a third mesh. A local target-response diagnostic accepts compliance discrepancy at most 10%, but this does not replace strict solid

agreement. Stress fields and volume-weighted 95th-percentile von Mises values are retained for inspection; compliance convergence does not certify stress convergence. Clamp singularities, small feature errors, yield, buckling, fatigue, contact and actual-use safety are outside this experiment.

Five of six case triples yield 15 solved source/target/prediction configurations with compliance mesh changes below 5%. Across their recorded meshes the largest normalized equilibrium/energy error is $8.20\text{e-}13$; final CAD-to-mesh volume error is at most 1.39%. Three predictions meet both strict geometry agreement and the local 10% target-compliance agreement tolerance. The propeller and flange predictions fail geometry while meeting that response tolerance. The screw has no FEM response because its boundary mesh fails; this also affects the source and released target, so it is not attributed solely to the model. The incorrect screw output contains seven CAD solids, versus one in its target. Boundary meshing encounters intersecting facets/segments; a source-only finer-mesh and alternative HXT-mesher diagnostic also fails. No component fusion, thread removal, contact model or geometry repair is used to manufacture a solution. This case remains not verified.

D. Multi-source Colab training pipeline

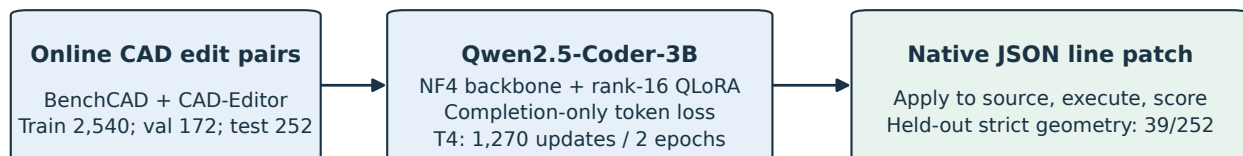
The reproducible extension uses paired edits from BenchCAD and Microsoft’s CAD-Editor in their native representations. The prepared split contains 2,540 training, 172 validation, and 252 test records: respectively 492/44/124 CadQuery pairs and 2,048/128/128 native sketch-extrude pairs. Two additional pipe-related held-out instructions are excluded from this extension; the earlier 126-case experiment is unchanged. Exact source/target connected components are isolated within each source. Cross-source geometric ancestry is not established. Token-length filtering is recorded before training and evaluation.

A fresh Qwen2.5-Coder-3B-Instruct backbone uses NF4 double quantization and completion-only QLoRA: rank 16, alpha 320, the seven attention/MLP projections in the last twelve blocks, two fixed epochs, seed 17, learning rate 10^{-4} , effective batch four, a 4,096-token context, and a 768-token output cap. The completed Colab T4 run uses microbatch one with four accumulation steps. Its saved manifest records `torch.bfloat16`; native BF16 hardware acceleration is not established. The final adapter is reloaded for greedy evaluation. Training completes 1,270 optimizer steps in 20,923.7 seconds with mean completion loss 0.2495. The adapter contains 9,977,856 trainable scalars; all 168 saved tensors are finite and their check-

Example	Source C	Target C	Prediction C	Discrepancy (%)	Combined diagnostic
Mounting angle	2.043e-06	7.484e-07	7.464e-07	0.259	Geometry + response agree
Locator block	5.185e-08	3.742e-08	3.742e-08	0.003	Geometry + response agree
Rotated shaft	5.287e-07	5.287e-07	5.291e-07	0.068	Geometry + response agree
Propeller	3.334e-07	5.242e-07	5.221e-07	0.409	Geometry fails; response agrees
Pan-head screw	—	—	—	—	Not verified: mesh failure
Round flange	1.208e-07	1.212e-07	1.212e-07	0.021	Geometry fails; response agrees

Table 5. Mechanical example FEM checks under explicit test scenarios. Compliance values use N mm. Discrepancy compares the prediction with the target, with independently generated meshes. A response match is not sufficient for a correct edit. Dashes denote failed meshing, not zero displacement or an invented solver pass.

CAD patch model: multi-source supervised training



FEM is an external post-edit check; its solver outputs do not supervise token training.

Figure 5. CAD editor architecture and supervised training. Paired native CAD edits train the quantized Qwen2.5-Coder-3B backbone with completion-only QLoRA. Patches are applied and independently scored on held-out geometry; FEM remains an external post-edit check rather than a training loss.

sum matches the run manifest.

The model maps an instruction and original native CAD to an original-coordinate JSON line patch. Representation-specific schema and syntax checks precede execution. CadQuery patches are compared to released STEP targets; CAD-Editor command patches execute through a six-bit sketch-extrude decoder, with locally reconstructed target solids explicitly distinguished from author-released STEP files. Geometry and failure counts remain separate by source. Only examples with explicit material, unit, fixture, and load contracts proceed to tetrahedral FEM. Equilibrium, energy identity, mesh-volume fidelity, and compliance sensitivity checks precede a combined geometry/response diagnostic. FEM is an external post-edit tool, not a language-generated pass label or a solver reward used in training. Invalid patches and failed meshes remain unverified. The notebook saves adapters, manifests, predictions, geometry metrics, and numerical fields before downloading an artifact bundle.

Completed held-out results. The final adapter obtains 39/252 strict matches (15.48%), versus 1/252 for the fresh base. Counts by source are 25/124 for BenchCAD and 14/128 for CAD-Editor. At $\text{IoU} \geq 0.95$,

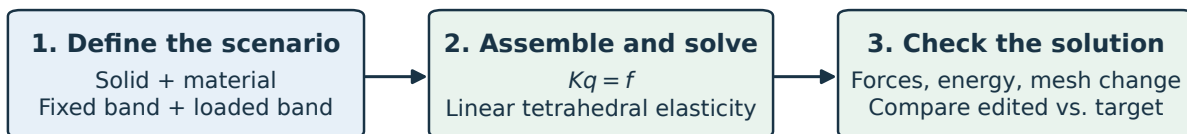
Model / source	n	Patch	Solid	Strict
Fresh 3B base, all	252	39	7	1
Final 3B adapter, all	252	173	139	39
Final, BenchCAD	124	91	65	25
Final, CAD-Editor	128	82	74	14

Table 6. Completed two-source Colab evaluation. Patch denotes applicable schema/syntax; solid denotes successful execution/comparison; strict denotes volume $\text{IoU} \geq 0.99999$. All retained tasks remain in n , including six reference errors and one final-model worker timeout. BenchCAD references are released STEP files; CAD-Editor targets are locally reconstructed native sequences.

the base and final adapter match 2 and 52 cases. Exact target-program/sequence agreement is 1 versus 32; exact patch agreement is 1 versus 31. No final generation reaches the output cap. The whole-worker timeout makes the scorer’s reference-valid bookkeeping model-dependent (246 base, 245 final); we therefore use the complete held-out denominator, not 245 as a reference-valid test set. Different datasets, model size and training settings prevent treating this run as an isolated improvement over the earlier 1.5B experiment.

FEM of the new Colab adapter. We attempt the same six illustrative held-out instructions with the new model’s unmodified patches. Four prediction/-target pairs—mounting angle, locator block, rotated shaft, and round flange—pass strict geometry and the scenario-defined numerical, mesh-volume and compliance checks. Their relative compliance discrepancies are respectively 0.001123 %, 0.046691 %, 0.009141 % and 0.013270 %. The flange is now a strict match; the older H100 flange in Fig. 9 remains an incorrect prediction from a different adapter. The new propeller patch fails syntax validation. The screw attempt leaves an incomplete pipeline directory; its resumed audit records failure, with no saved FEM solution establishing an original cause. Both cases remain unverified. Saved meshes and displacement/stress fields substantiate the four passes; source fields are from the separate earlier audit, not recomputed for this pairwise Colab audit. These are diagnostic test scenarios, not operating-safety evidence or full-benchmark FEM coverage.

FEM: from an edited CAD solid to a checked response



Blue: fixed nodes Red arrow: resultant load direction All surfaces are undeformed.

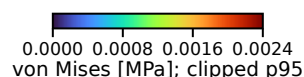
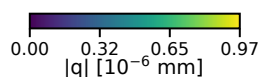
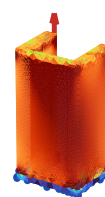
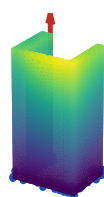
Mounting-angle width edit | actual final 3B prediction

33,871 nodes; 142,334 tets | $C = 7.461e-07$ N mm | $\max |q| = 9.750e-07$ mm

Tetrahedral surface mesh

Displacement magnitude

Diagnostic stress field



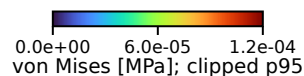
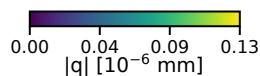
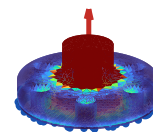
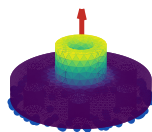
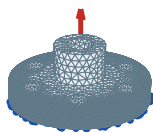
Flange: 4-to-6 mounting-hole edit | actual final 3B prediction

15,035 nodes; 59,326 tets | $C = 1.214e-07$ N mm | $\max |q| = 1.331e-07$ mm

Tetrahedral surface mesh

Displacement magnitude

Diagnostic stress field



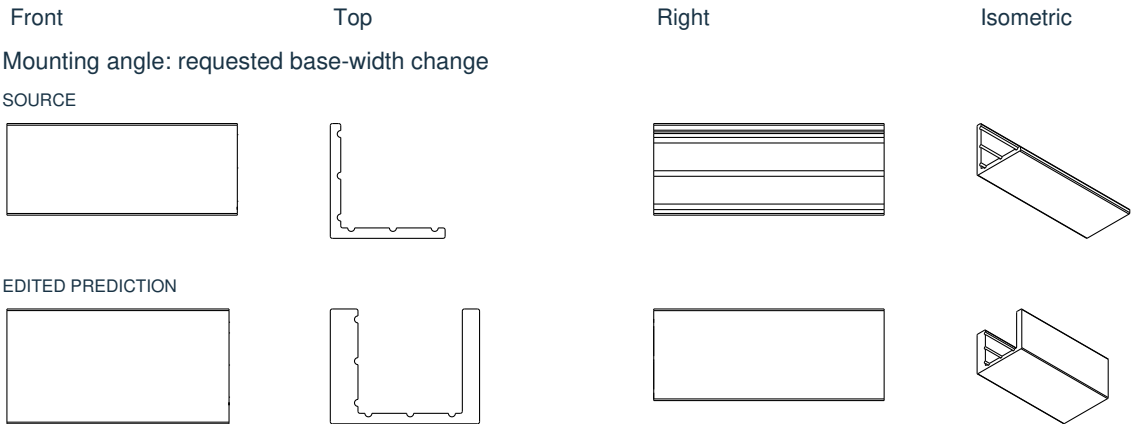
Worked-example interpretation

The same solid supplies the mesh and both field plots. A color change is a solver response, not proof that the edit is correct: independent solid and target-response checks are still required.

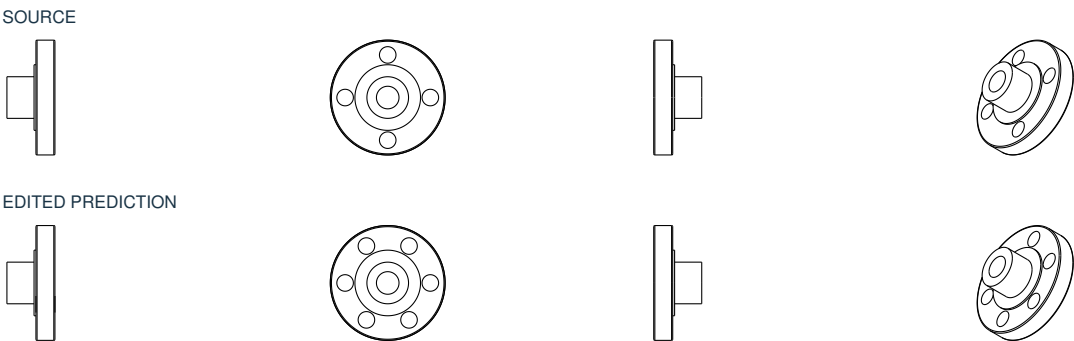
Assumed $E = 210,000$ MPa, $\nu = 0.3$; static 1 N test. Stress convergence / operating safety are not established. Surface edges display the tetrahedral boundary, not every interior element.

Figure 6. Large FEM method walkthrough with actual mechanical edit examples. The final 3B mounting-angle and six-hole flange predictions supply all meshes and fields. The left column displays the tetrahedral boundary mesh; the middle shows displacement magnitude; the right shows diagnostic von Mises stress, clipped at the per-example 95th percentile. Blue nodes are fixed and red arrows show the resultant load direction. Independent CAD/reference agreement, equilibrium, energy and mesh checks accompany these plots; stress color alone establishes neither correct editing nor strength. The listed values use each saved fine mesh and the disclosed synthetic static scenario.

Actual CAD-derived drawings: source and final 3B edited prediction

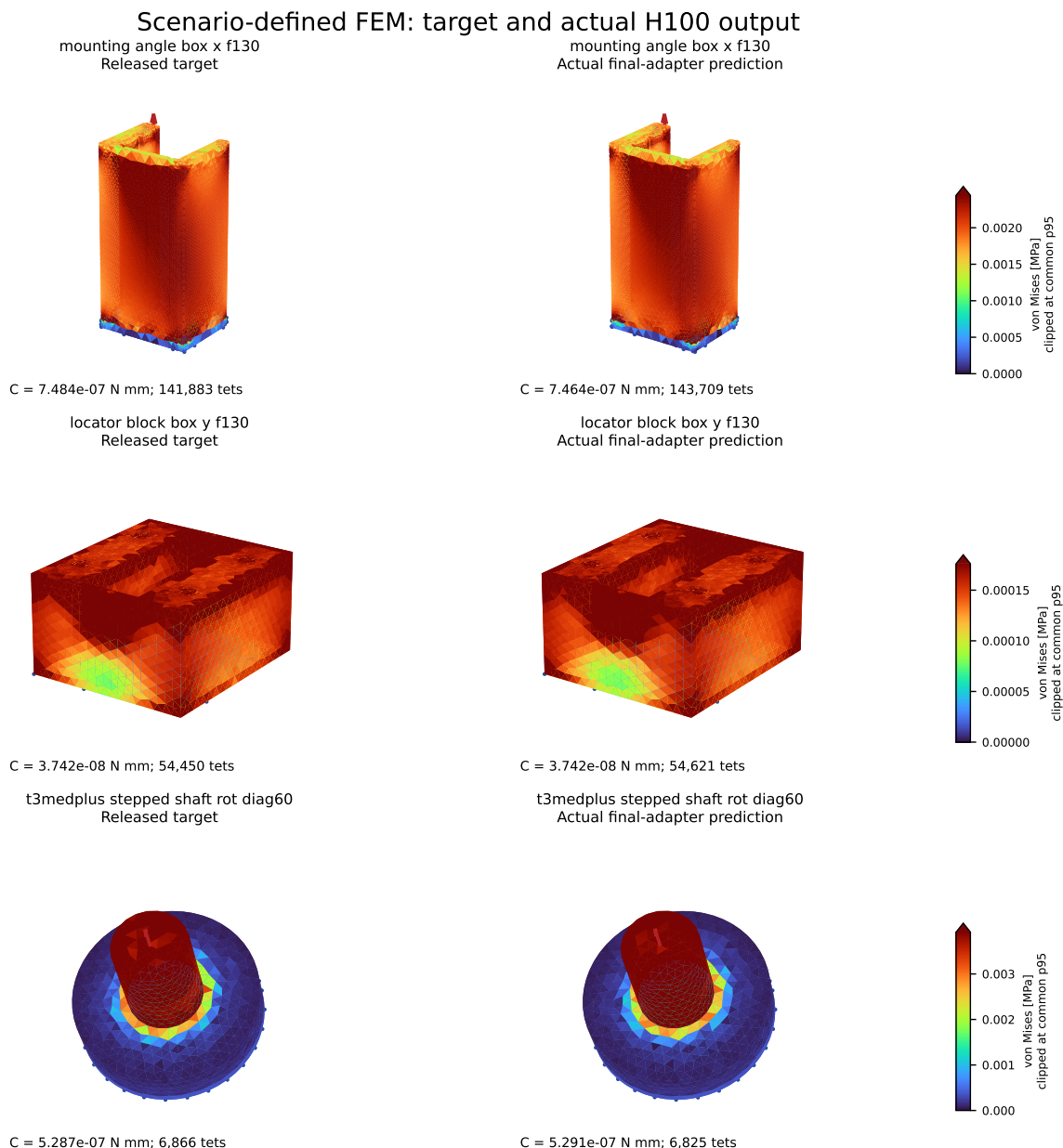


Flange: 4 to 6 mounting holes; same diameter and bolt-circle radius



Projected views are autoscaled; dimensions and response must be read from CAD / solver artifacts.

Figure 7. Linked engineering drawings from actual mechanical CAD solids. Front, top, right and isometric views show the original and final 3B predicted mounting angle and flange. The flange edit changes four mounting holes to six equally spaced holes while retaining their diameter and bolt-circle radius; each panel is independently autoscaled. Both edited predictions pass the disclosed static geometry/FEM diagnostics. The views contain no inferred stress, production dimensions or hidden geometry repair.



Assumed $E = 210,000$ MPa, $\nu = 0.3$; 1 N test load. Blue: fixed nodes; red: load direction.
 Undeformed meshes; compliance mesh check only. Stress fields are diagnostic, not a strength/safety verdict.

Figure 8. Actual FEM fields for three geometry-matching H100 edits. Released targets and model outputs use the documented synthetic test contracts. Blue points identify fixed nodes; arrows indicate the load direction. Undeformed surface stresses share a scale within each row, clipped above the common 95th-percentile value. Compliance passes the mesh-sensitivity check; stress colors are diagnostic rather than a strength verdict.

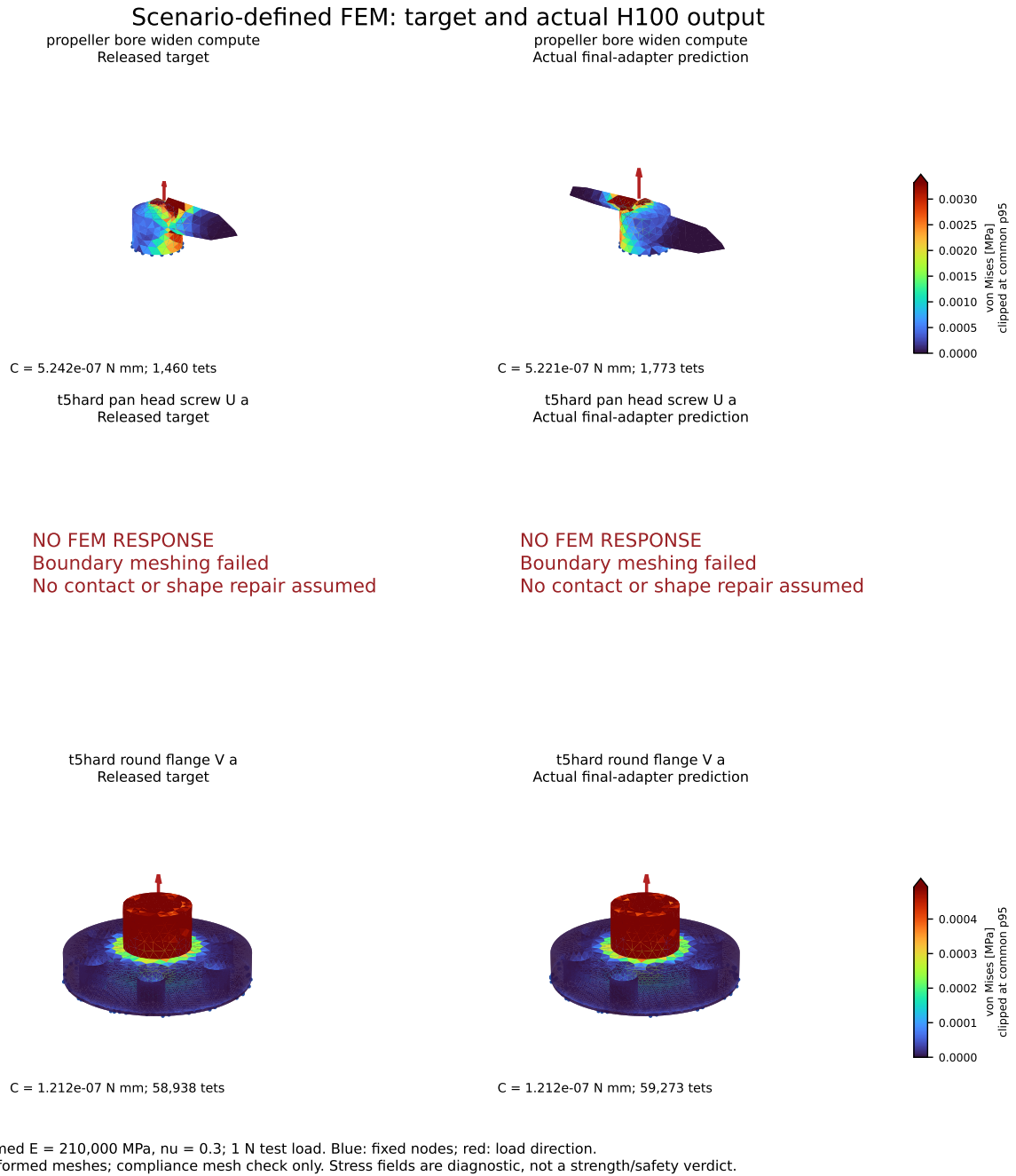


Figure 9. FEM diagnostics do not replace edit fidelity. Incorrect propeller and flange shapes produce similar compliance under the chosen test load, but fail strict reference geometry. The screw row has no FEM field because meshing fails; blank panels are explicitly marked. The same stress-scale and fixture conventions apply as in Fig. 8.